

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda_s \mathcal{L}_{\text{slope}} + \lambda_p \mathcal{L}_{\text{peak}}, \quad \lambda_s = 0.5, \quad \lambda_p = 0.05$$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{BF} \sum_{b=1}^B \sum_{f=1}^F (\hat{y}_{b,f} - y_{b,f})^2$$

$$\mathcal{L}_{\text{slope}} = \frac{1}{B(F-1)} \sum_{b=1}^B \sum_{f=1}^{F-1} (\Delta \hat{y}_{b,f} - \Delta y_{b,f})^2, \quad \Delta y_{b,f} = y_{b,f+1} - y_{b,f}$$

$$\mathcal{L}_{\text{peak}} = \frac{1}{B} \sum_{b=1}^B \sum_{f \in \mathcal{P}(y_b)} (\hat{y}_{b,f} - y_{b,f})^2, \quad \mathcal{P}(y_b) = \text{local maxima of the true spectrum}$$